Integrated project execution and management package

ONGC Mumbai High Bridge Refurbishment & Replacement Project | Tender ZW1PC26002 | Shivansh |25 October 2026 | P14 Rev 0

Issue purpose: management development and review. This issue consolidates requirements, accountable workstreams, operational registers and a reproducible schedule scenario. It does not certify completed structural design, executed procurement/construction, approved AFC drawings or an accepted contract baseline. Actual commencement, resource commitments, productivity, quotations, progress and approvals remain unverified.

Executive decision brief

The source set supports developing a disciplined EPC control system. It does not establish a fully funded and resource-proven 12 month execution plan. The previous 207 activity XML had 124 links, 83 activities without predecessors, 83 without successors and 17 isolated tasks, and contained no resource assignments. The supplied Gantt image does not supply enough activity/logic detail to validate it.

P14 provides 285 linked activities with explicit calendars and proposed resource queues. Using an assumed 1November 2026 start, sequential engineering reviews, 45 calendar-day steel lead times, two fabrication bays at 4MT/productive bay-day, one coating cell and the stated offshore-season planning exclusion, calculated completion is 6April 2028. This is 157 calendar days beyond the assumed 1November 2027 anniversary, before considering any qualifying contractual relief. It is not an actual delay/LD determination.

The driving chain passes through survey/basis acceptance, BHS analysis/detail approvals, steel supply, fabrication-bay queues, coating queues, the final strengthening package, offshore service/removal/punch work and turnover. Additional fabrication capacity alone moves the scenario to 26February 2028; reducing steel lead time as well does not improve that finish under these other constraints. Review logic, first releases, coating/service capacity and seasonal workability must be addressed together.

Immediate management decisions: confirm actual contract and governing revisions; freeze survey/design scope; resolve quantity and bearing/load queries; obtain evidenced supplier/yard/crew/vessel capacities; price and fund the full lifecycle; validate and approve a feasible completion/recovery strategy. No claim of outperforming Saipem/McDermott is supported. The quality standard here is traceable evidence, explicit gates, realistic dates and measurable delivery performance.

1. Source coverage and document precedence

Latest batch: 68 uploads; 67 opened; 52 distinct contents; 7 distinct contents have no exact prior local hash match. New contents include chart.pdf, Stage 27/30/31/32 workbooks and distinct Volume 2/3 contents. The large preliminary PDF exceeded 32MiB and was not reviewed. P13 previously opened 76 files with 21 distinct contents. A full file-by-file inventory is in the workbook; repeated bytes reuse earlier analysis and retain its limitations. This is not a claim that every table and historical attachment has received exhaustive numerical/contract review.

chart.pdf is an image-only Gantt-like graphic at poor readable scale; it provides no extractable activity network. ticket.pdf from P13 is a travel booking; no project requirement or actual mobilisation/award was inferred, and personal booking details are excluded. The ZIP master-plan archive was opened/integrity-checked and member hashes indexed; archive contents do not establish engineering completion.

BRRP-specific sources govern subject to the contract hierarchy and amendments. Sections 1–3 titled Sump Caisson and SWLP Casing Replacement Project require explicit incorporation/applicability before adoption. The latest supplied extension notices end at 23June 2026 submission/24 June 2026 unpriced opening; later notices and actual opening remain unconfirmed. SCC copies and A3 are identical to earlier files; the announced revised versions are not positively identified. SVL12.3 was located in Combined 3 pp 261–341 and replaces the earlier missing-list issue, subject to amendment and vendor-status checks.

Source hierarchy is to be reconciled using B-4 and SCC1.2; SCC overrides GCC for their conflict. Do not adopt a bidder’s proposed hierarchy or concession from a query table when the company reply does not accept it. Preserve original source files and dated issued revisions; record deletions and amendment overlays rather than editing historical originals.

2. Bridge scope and quantity baseline

ICW-ICG UD | Strengthening | 49 | 25

ICW-ICG LD | New/replacement | 122 | 1

ICG-ICP UD | Strengthening | 116 | 25

ICG-ICP LD | New/replacement | 160 | 1

BHS-SLQ | Replacement/services | 282 | 210

SLQ-WIS | Strengthening | 91 | 15

NQO-NQD | Strengthening | 93 | 25

WIN-NC | Strengthening | 46 | 10

SCA-SC1 | Strengthening | 63 | 15

SCA-SCF | Strengthening | 119 | 30

Source-row construction totals 1141MT versus printed 1142MT, including landing-strengthening 216 versus 217MT. Handrail 1166 versus 1168 m and grating 718 versus 719 m² also remain unresolved; demolition 357MT and GI1274 m² sum consistently. These are tender transcriptions, not released member/plate purchase quantities or verified module weights. MTO quantities must be rebuilt from accepted drawings and measured condition before procurement.

Three replacement/new packages are ICW–ICG lower deck, ICP–ICG lower deck and BHS–SLQ. Seven strengthening packages retain their bridge-specific conditions and service interfaces. Upper/lower deck labels and NQO/NOQ nomenclature must be controlled. BHS demolition is a distinct engineered operation after the accepted service/support transition, not an assumed concurrent steel-removal allowance.

3. Engineering basis, deliverables and release

Design life — New bridges 25 years. Source: CriteriaPart II p 119. Confirm remaining/new component applicability.

Elastic modulus — 210000 MPa. Source: Criteria p 66. Use approved material model.

Density — 7850 kg/m 3. Source: Criteria p 66. Coatings/attachments/rigging separate.

Global walkway live load — 250 kg/m 2. Source: Criteria pp 42–43. Do not replace local grating criteria.

Local grating load — 500 kg/m 2. Source: Criteria p 70. Apply local design/approved combinations.

Clear walkway — At least 1.2 m plus services allowance. Source: Criteria pp 42–43. Verify actual service envelope.

Piping load allowance — Operating/planned future plus 20% as stated. Source: Criteria pp 42–43. Resolve double-counting of future services.

Bearing system — Pinned/hinged and sliding/PTFE-equivalent with guides. Source: Criteria pp 42–43. Resolve BHS fixed/sliding source conflict.

Movement/thrust — 150% specified platform translations/thrust basis. Source: Criteria pp 42–43. Use verified platform movements and reactions.

Deflection — Vertical L/400; lateral wind L/500. Source: Criteria p 86. Verify governing span and load case.

Camber — Compensate DL; studies and generic L/200 conflict. Source: Criteria/source bridge studies. Resolve profile rather than assume L/200.

Weight/COG reanalysis — Weight>5%; COG>2% corresponding dimension. Source: Criteria p 20. Track engineered thresholds and actual modules.

Strengthening sequence — ICG/ICP/BHS wrap additions before new placement. Source: Structural SOW addendum p 1. Engineer connection/stability; raw shell mass not capacity proof.

Piping movement — Differential platform movement/extreme storm flexibility. Source: Combined 3 p 260 clause 5.3. Iterate stress/support/bearing reactions.

Geometry reference — Tender dimensions indicative; survey required. Source: Combined 1 GA notes; A9. Do not adopt study span as surveyed geometry.

The Engineering DCI register contains 170 proposed bridge deliverables plus 13 platform assessment entries. IDs prefixed SHV are proposed management IDs, not issued source drawing numbers. Combined 1Annexure 1 calls for global deck assessments ICP/ICG/ICW/BHS/SLQ and local checks WIS/NOQ/NQD/WIN/NC/SC1/SCA/SCF. Project applicability and naming need confirmation. Fifteen specific technical queries are recorded with source, owner and affected release in the Technical Queries sheet. The source new-bridge GAs are recorded separately; source indicative dimensions and source study utilisation ratios cannot substitute for checked project design.

C8 document categoriesA/I/R differ from review dispositionsA/B/C. Mother-document submission/resubmission/approval statesS/R/A must be mapped per applicable row. General review is 10 working days, with longer listed first reviews for specified documents; subsequent reviews and MCI stages require their stated treatment. Stage-specific project DCI/MCI and submission staggering are essential. The P14 scenario uses 10 working-day allowances for selected bridge-review activities, without proving every comment/review iteration is included.

Analysis must cover operating and temporary stages, load combinations, bridge/platform displacement, member/joint capacity, fatigue/VIV where applicable, bearing travel/thrust/temperature setting, landing/local deck adequacy and lift/transport/removal stages. Detail welds, bolts, tolerances, camber, corrosion protection and inspection access from accepted calculations. Independent checking, CRS closure, correct source revision and authorised release are mandatory gate evidence. No approved AFC drawing sheets are created by a register.

4. A–Z execution procedures

A — Accountability and authority

Accountable role: Project Director. Inputs: NOA/contract, scope split and corporate delegation.

Procedure: Appoint named managers and deputies; record availability and competing commitments; establish RACI; set spending/change/technical-release authority; chair weekly integration review.

Acceptance evidence: Signed organisation/RACI and delegated-authority register.

Control: Named accountable owner for each control; no unidentified release authority. Source: SCC1; Technical BEC5.2–5.4.

B — Baseline and requirements

Accountable role: Contracts Manager. Inputs: Tender, addenda/corrigenda, source drawing register.

Procedure: Maintain clause-to-deliverable matrix; resolve SCC/GCC precedence using B-4; link revisions and deletions; isolate S&SRP references; freeze controlled scope by bridge and retain open discrepancies.

Acceptance evidence: Approved requirements and scope baseline; assumption/change log.

Control: Every requirement has source, owner, verification and disposition. Source: P8–P12; CQ replies; B-4.

C — Contract, qualification and compliance

Accountable role: Contracts Manager. Inputs: Actual bidding route, experience evidence and amendments.

Procedure: Check 110MT/marine route and consortium exceptions; retain PM/qualifying main activity; obtain subcontract commitments; verify TPI authentication and valid accreditation; reconcile financial evidence; manage securities and integrity-pact disclosures.

Acceptance evidence: Qualification/compliance matrix and evidence bundle.

Control: No unsupported bidder confirmations or expired verification evidence. Source: Technical BEC5; P9/P11.

D — Design basis and survey

Accountable role: Engineering Manager. Inputs: BRRP criteria, tender GAs and survey access.

Procedure: Verify coordinates, elevations, spans, clearances, member sizes/thickness/loss, support movement, piping/cable loads and NDT; create controlled datums; close bearing/camber/load discrepancies; record later company instructions under change control.

Acceptance evidence: Accepted survey dossier and bridge-specific design basis.

Control: Survey-derived geometry and load status visible on every engineering package. Source: CriteriaPart II; Corrigendum 3 GCC2.3.6.

E — Engineering and independent checking

Accountable role: Engineering Manager. Inputs: Accepted survey, criteria, scope and discipline interfaces.

Procedure: Analyse bridge and supporting platforms, members/joints, bearings, fatigue/VIV, preservice lifts/transport and temporary stages; iterate piping reactions; detail connections; perform independent check and comment closure; submit through controlled DCI.

Acceptance evidence: Checked calculations, interface reactions, design reports and accepted drawings.

Control: No source-study utilisation ratio represented as new design proof. Source: C8; Combined 1Annexure 1; DesignCriteria.

F — Fabrication drawings and AFC release

Accountable role: Lead Structural Engineer. Inputs: Accepted mother analysis and verified geometry.

Procedure: Issue source-to-detail mapping, shop GAs, member/cut lists, joint/weld maps, bearing/guides, padeyes/supports and temporary-stability details; apply source revisions and CRS; independent check then company/CA release as applicable.

Acceptance evidence: Signed bridge-specific release index with matching drawing/MTO/procedure revisions.

Control: AFC status requires actual authorised approval and field-copy control. Source: C8 mother documents; Combined 1 pp 27–41.

G — Geometry, MTO and weight control

Accountable role: Quantity/Weight Engineer. Inputs: Released model/drawings and field verification.

Procedure: Separate new bridge, strengthening, new landing, existing landing, demolition and finish quantities; resolve 1MT/2 m/1 m² discrepancies; calculate member-length/plate/bolt/weld/coating MTO and wastage explicitly; track module dry/wet/rigging weights and COG.

Acceptance evidence: Traceable MTO revisions, reconciliation and verified module weight report.

Control: No bulk tonnage substituted for lift weight, member BOM or coating area. Source: SOW quantity table; MTOIRT; Criteria weight/COG trigger.

H — HSE and SIMOPS

Accountable role: HSE Manager. Inputs: Workfront hazards, operating platform and vessel requirements.

Procedure: Complete HIRA/JSA/PTW, isolation/LOTO, hot work, gas testing, work-at-height/rescue, dropped objects, scaffold/access, marine transfers, occupational exposure and emergency interfaces; train staff; issue certified PPE and validate work-start clearance.

Acceptance evidence: Approved HSE/SIMOPS plan, personnel matrix and permit packs.

Control: Zero work starts without applicable permits, isolation and readiness evidence. Source: SCC7/8; PPElist; OISD155 applicability.

I — Interface and operations management

Accountable role: Interface Manager. Inputs: Platform operations, engineering and contractors.

Procedure: Register structural/piping/electrical/instrument/telecom/firewater interfaces; agree outage and transfer sequence; assign tie-in owners; confirm access and emergency routes; track support reactions and dimensional tolerances; resolve NQO/NOQ naming.

Acceptance evidence: Signed interface agreements, outage windows and battery-limit register.

Control: No unowned interface on lookahead or lift plan. Source: SCC1.7; Combined 1/2/3.

J — Joint inspections and quality assurance

Accountable role: QA/QC Manager. Inputs: Project quality plan, codes, specifications and approved ITP.

Procedure: Set inspection/hold/witness points; qualify welding and personnel; control heat/consumable traceability; calibrate instruments; inspect fit-up/weld/NDT/coating/dimensions; repair and close NCRs; issue releases and indexed records.

Acceptance evidence: ITP, qualification dossiers, inspection releases and NCR closure.

Control: No concealed work or next-stage release through an unresolved hold point. Source: FS2009F; FS2005Rev 9; C8.

K — Key suppliers and procurement

Accountable role: Procurement Manager. Inputs: Approved requisition, MTO, PS and compliant sources.

Procedure: Package 25 procurement scopes; RFQ and technical/commercial evaluation; verify SVL12.3 status/product/location, steel melt/pour/DVA and traceability; obtain PS approvals, place authorised PO, expedite and inspect; preserve vendor-data dependencies.

Acceptance evidence: PO/delivery/inspection register with quoted dates and evidence.

Control: No unpriced allowance treated as commitment; no silent substitution. Source: SVL12.3 Combined 3 pp 261–341; Corr 3; C8.

L — Logistics, loadout and preservation

Accountable role: Logistics Manager. Inputs: Released modules, lifting/transport studies and cargo slots.

Procedure: Verify shore-base storage/load capacity, route permits, lifting gear and packing/preservation; produce cargo ID/weight/COG/manifests; sea fasten and inspect loadout; monitor voyage/receipt and backload; segregate quarantine and accepted stock.

Acceptance evidence: Approved loadout/transport pack, manifests and receipt reports.

Control: Cargo release ties to exact module/drawing/weight revision. Source: SCC14; C9/C10; applicable marine assurance.

M — Marine engineering and assurance

Accountable role: Marine Manager. Inputs: Module envelope, vessel particulars and environmental data.

Procedure: Verify actual-radius crane charts, dynamics/rigging/padeyes, deck strength/stability, seafastening/voyage, positioning/mooring/DP, subsea hazards, access/POB and certificates; obtain applicable MWS/company clearances and predeployment audit.

Acceptance evidence: Accepted vessel/operation assurance pack and go/no-go record.

Control: SCC5 tonne transfer crane is not a bridge-lift selection criterion. Source: SCC3/5; C9/C10; C8 starred requirements.

N — Network schedule and calendars

Accountable role: Planning Manager. Inputs: Scope/quantities, approvals, capacity and supplier dates.

Procedure: Use one logic network with source/assumption status; model office/yard/offshore calendars, approval states and PS stages; link shared resources and completion dossiers; calculate forward/backward paths, target shortfalls and sensitivities; freeze only after validation.

Acceptance evidence: Approved baseline and update narrative with calculation settings.

Control: No day-number bar chart or register labelled approved CPM. Source: SCC2/11; C8; P14 scenario.

O — Offshore installation and temporary stability

Accountable role: Construction Manager. Inputs: Approved IWP, design, loadout and operating windows.

Procedure: Complete preconstruction survey/access; strengthen supports before new bridge placement where required; verify temporary load paths and restraint removal order; authorise lifts, landing alignment/bearing travel and permanent connection; inspect before transfer and removal.

Acceptance evidence: Signed workfront release, installation survey and inspection dossier.

Control: No bridge removal until alternate support/service sequence accepted. Source: Structural SOW addendum; bridge-specific approved methods.

P — Production and fabrication control

Accountable role: Fabrication Manager. Inputs: Released shop packs, material, qualified personnel and capacity.

Procedure: Validate bay dimensions/cranes and shifts; material receipt/heat mapping; cut/form/fit/weld with distortion/camber controls; NDT/repair; dimensional survey/trial fit; coating/curing; preserve and release modules by accepted quantities.

Acceptance evidence: Daily accepted output, resource hours and module release.

Control: Separate installed structural MT from purchased stock and rework. Source: MasterPlanV3; FS2009F/2005.

Q — Quantity, progress and earned value

Accountable role: Project Controls Manager. Inputs: Approved work quantities, budget and progress rules.

Procedure: Measure installed and accepted quantities by workpack; distinguish fabrication/install completion, certification, invoice and cash; apply C8 engineering/MCI status weights only to their contractual bases; retain e-MB/SES traceability; do not earn budget twice.

Acceptance evidence: Signed measurement, progress-rule and cost-code reconciliation.

Control: EV/SPI/CPI blank where BAC/PV/AC or accepted quantities absent. Source: SCC12; C8 review/progress rules.

R — Risk, contingency and recovery

Accountable role: Risk/Planning Manager. Inputs: Risk register, driving path and resource limits.

Procedure: Identify causes/events/effects; score ownership and early-warning; quantify actual schedule/cost exposures when data exist; test material delays, re-reviews, vessel failure and workability; price recovery options and seek authorised decisions.

Acceptance evidence: Owned mitigations, contingency and recovery decision log.

Control: No probability or P80 claim without a risk model and evidence. Source: SCC11; P14 alternatives.

S — Subcontracts and retained responsibilities

Accountable role: Contracts Manager. Inputs: BEC route, agency scope and signed commitments.

Procedure: Define battery limits and deliverables; retain required PM/main activities; obtain capacity/commitment letters and flow down QA/HSE/integrity obligations; manage approved agencies, claims and performance; prohibit unauthorised whole-works pass-through.

Acceptance evidence: Approved responsibility split and signed subcontract evidence.

Control: Agency listing alone is not signed capacity or eligibility. Source: A16/A17/B6; Technical BEC5.

T — Treasury, cost and cash flow

Accountable role: Finance Manager. Inputs: Priced estimate, signed terms and actual milestones.

Procedure: Track budget/commitment/accrual/actual/ETC/EAC; schedule taxes, wages, charters, vendor payments, BG margins/fees/releases, certification/payment lags, retention/advance recovery and financing; extend through final account/warranty releases.

Acceptance evidence: Priced control budget and funded cash forecast.

Control: Unknown inputs remain unknown; no zero-cost feasibility conclusion. Source: P9/P11 securities; A3/A10; SCC.

U — Utilities and service reinstatement

Accountable role: Commissioning Manager. Inputs: Approved tie-ins, isolations and test requirements.

Procedure: List every piping/cable/instrument/lighting/earthing/firewater service; verify temporary bypasses; isolate, transfer, inspect/test and restore under operations control; close pressure/leak, continuity/IR/earthing and functional tests as applicable.

Acceptance evidence: System-wise reinstatement/test sheets and operations acceptance.

Control: No energisation or hydrotest without approved test boundary and procedure. Source: Combined 1/2/3 discipline specifications.

V — Variations, delay and claims records

Accountable role: Contracts Manager. Inputs: Written instruction, baseline and contemporaneous records.

Procedure: Log notices and contractual time bars; maintain monthly joint critical-event logs and quarterly delay reports; substantiate quantities/invoices/man/barge days; apply SCC9 positive/negative valuation and avoid duplicate fees; document relief decisions.

Acceptance evidence: Notices, joint statements, approved variations and final-account evidence.

Control: No assumed entitlement from a schedule overrun alone. Source: SCC9/11; Corr 3 clause 2.3.6; GCC8.

W — Weather and workability

Accountable role: Marine/Planning Manager. Inputs: Field metocean, forecast and vessel limits.

Procedure: Set engineered operating/abort limits by operation; agree forecast/lookback records and weather standby attribution; plan refuge/demob/remob and emergency triggers; preserve contemporaneous proof for qualifying monsoon relief.

Acceptance evidence: Weather logs, operational decisions and seasonal resource plan.

Control: Planning blackout is not automatic contractual extension. Source: SCC2/4; C9/C10.

X — Execution readiness and daily control

Accountable role: Construction Manager. Inputs: IWP, crew/tools, quantities, materials, permits and assurance.

Procedure: Run 6 week constraint removal, 14 day lookahead and daily permit/readiness meetings; verify all gates and evidence IDs; record planned/accepted output and crew-hours; escalate blockers and resource clashes without bypassing safety gates.

Acceptance evidence: Released daily work list, production/constraint records.

Control: Readiness requires evidence and verified scope, not eight unchecked Yes labels. Source: Stages 30–35; HSE/QA project controls.

Y — Yard-to-field completion and handover

Accountable role: Completion Manager. Inputs: ITRs, drawings, tests, systems and punch categories.

Procedure: Create bridge/system turnover boundaries at design stage; collect manufacturing records progressively; inspect structural/bearing alignment and reinstated services; close blocking punch items; deliver as-builts/native files/vendor dossiers and scrap receipts; obtain formal acceptance.

Acceptance evidence: Accepted turnover dossier and handover certificate.

Control: No completion earned solely by physical steel placement. Source: A10; C8; SCC14; GCC completion clauses.

Z — Warranty, final account and lessons

Accountable role: Project Director. Inputs: Acceptance terms, outstanding obligations and final payments.

Procedure: Track defect liability/warranty and insurance/security releases; resolve claims and subcontract final accounts; retain integrity-pact/record obligations; archive controlled native/PDF data and access; close lessons and verify promised performance against outcomes.

Acceptance evidence: Contract closure certificate, release schedule and retained records.

Control: Physical completion, financial closure and obligation expiry are separate dates. Source: IntegrityPact 9; GCC/SCC governing periods.

5. Workpack and daily production system

The workbook defines 120 bridge/task workpacks across survey, engineering, material, fabrication, coating, loadout, supports, installation, service transfer, removal, punch and turnover. Each workpack requires verified quantities and a controlled evidence index. An applicability disposition is required when a generic task category does not apply. The screen returns HOLD until all eight listed gates are YES; even then it returns VERIFY EVIDENCE, not an authorised release. Only the designated authority can release accepted work.

Six-week review removes constraints; 14-day lookahead confirms material, access and capacity; daily meeting checks PTW, weather/SIMOPS and named workfront readiness. Log actual accepted quantities, crew-hours, nonproductive causes and rework against the IWP. Weekly forecast uses accepted output/productive crew-day only after sufficient comparable data exist. Separate productivity changes from changed scope and rework; do not divide aggregate steel tonnage by headcount and call it a verified installation duration.

6. Procurement, fabrication and marine execution

Twenty-five proposed procurement packages consist of ten bridge-structural scopes and fifteen common materials/services. Each requires applicability, quantities/specification, PS/vendor approvals, price, dates, inspection and preservation. P14 does not issue any PO or book a vessel. Validate SVL12.3 product/location/status; reference-only list amendments do not erase bans or B3 restrictions. Apply Corrigendum 3 Melt&Pour/DVA and Form 1 requirements by policy classification, not blanket assumptions about every item.

Fabrication sequence: released shop pack and heat-traceable material → qualified personnel/process → cut/fit-up and temporary stability/distortion control → weld/NDT/repair closure → dimensional survey/trial fit → surface preparation/coating/curing → verified module weight/COG and preservation → independent release for loadout. A shared coating cell is explicit in the scenario; actual area, cure system and physical cell capacity must be verified. Yard bays are assumed available for new steel after the prior package’s welding phase; retained bay occupancy and NDT constraints require validation.

Marine sequence: engineered module envelope → suitable vessel/actual-radius chart/deck/stability/positioning → cargo support and sea-fastening/voyage assurance → accepted loadout/certificates/audit → forecast/workfront go/no-go → transport and offshore receipt → pre-install support strengthening → lift and bearing landing/connection → accepted service transfer → engineered removal/backload → as-built survey and punch. Plan ICG/ICP/BHSwrap strengthening before new placement as required. Transfer-crane minimum 5 tonnes is not proof of bridge-lift capacity; minimum 60 special-personnel accommodation is not total project POB validation.

7. Schedule method, results and recovery

The schedule is a development-level integrated network, not a detailed constructability-validated L4 baseline. All 285 activities have unique IDs and valid predecessors, the graph is acyclic and forward/backward passes obey FS0. Explicit resource queues cover two fabrication bays, one coating cell, three strengthening crews, one heavy-lift chain, one cargo chain, one service-transfer team and one removal team. Other capacities, shared platform access, POB and engineering/NDT labour are not fully modelled.

Office work usesMonday–Friday without holidays; yard usesMonday–Saturday; offshore uses 7 days outside 16May–15 October and treats each offshore phase as indivisible within an open season. This is a planning exclusion, not an automatic contractual time extension. Vendor deliveries usecalendar days. Dates are half-open:

finish is the next boundary after productive work; milestone dates lie on that boundary. Float is calendar-day latest-start slack to scenario completion, not approved contract float.

BASE | 2bays×4MT/day;45day steel supply | 2028-04-06

MORE_YARD | 4bays×8MT/day;45day steel supply | 2028-02-26

FAST_SUPPLY_MORE_YARD | 4bays×8MT/day;15day steel supply | 2028-02-26

All three scenarios preserve review allowances; none proves the 12 month target. The common sequence waits for campaign survey/basis then completes staged analysis/detail approvals before steel PO; this is deliberately conservative and must be tested against lawful independent packages and verified survey rolling releases. Ten design streams run in parallel without a verified staffing model, so the scenario is also optimistic about engineering capacity. It is neither a guaranteed upper bound nor lower bound.

Recovery options to evaluate: independent early survey/design release boundaries; approved overlap in design preparation while respecting mother-document review states; compliant stock and supplier reservations with explicit commercial authority; validated higher bay/coating/NDT throughput; module/transport strategy and operations windows; separate service/strengthening crews where workfront/POB permits; alternate seasonal execution with contractual notice/relief. Price each option and rerun full dependencies. Do not compress mandatory inspections, cure times or review periods merely to achieve a date. No procurement or at-risk fabrication is authorised by this report.

8. Cost, cash and progress

The control workbook records 25 cost accounts and 156 weekly cash periods, extendable to actual obligation expiry. Budgets, rates and cash inputs intentionally remain blank. These are operating templates, not a priced estimate or funded forecast. Cost EAC is actual paid plus all remaining expenditure, including accrued unpaid/commitments/uncommitted ETC once, with no duplication. Cash separates receipts, retention/security releases, vendor/labour/marine/tax-finance and other flows. Weekly net cash is withheld until all eight input categories are numeric and the row is marked VERIFIED; enter explicit zero only for evidenced nil cash. Cumulative cash is withheld if any preceding week is incomplete. Opening cash must be funded and evidenced.

Use amended A10 milestones, not arbitrary revenue percentages. Earlier amendment deletes separateMob/Demobitems 12/13 and revises repair-arrival 20 to 25%; uploaded base MPF still contains original lines. Reconcile the entire allocation before forecasting. Bid security extends 45 days beyond applicable bid validity; validity anchor and supporting-company guarantee dates follow corrigenda. Guarantee face value differs from margin cash and bank fees. Do not assume mobilisation advance, tax exemption, retention release or certificate/payment lags without terms.

Engineering review dispositions and MCI stage weights apply only to their prescribed bases. Accepted physical work, certification, invoice, net collection and EV are separate measures. SPI/CPI require approved PV/BAC/AC; no meaningful ratio is available from zero/missing inputs. Daily e-MBmeasurements and approved SESlinks are billing prerequisites under SCC12. Monthly critical-event logs and quarterly joint delay analysis are deliverables; the workbooks do not upload them automatically.

9. Risk, assurance and completion

R01 — Wrong source revision or unrelated S&SRP scope. Warning: Revised SCC/A3 not positively mapped; common platform names. Owner: Contracts. Response: Freeze affected release and resolve applicability/precedence. Gate: PO/AFC release blocked for disputed requirement.

R02 — Survey geometry or corrosion differs from tender. Warning: Span/elevation/thickness/support condition outside accepted model. Owner: Engineering. Response: Survey, NDT, independent check and controlled redesign. Gate: Affected design and load-path release held.

R03 — Steel grade/origin unavailable in lead time. Warning: Mill quote fails grade, dimensions or Melt&Pour/DVA evidence. Owner: Procurement. Response: Qualify alternative compliant mill/section through design approval. Gate: No unauthorised substitution.

R04 — First offshore season missed. Warning: Ready-for-installation forecast beyond 15May under scenario. Owner: Planning. Response: Stage independent releases and verify earlier supply/capacity; cost second season.

Gate: Seek schedule/contract decision with evidence.

R05 — Vessel capacity or availability fails. Warning: Crane chart, deck/stability or booking does not cover module. Owner: Marine. Response: Engineer module envelope and qualify alternate spread. Gate: No charter/lift commitment without assurance.

R06 — Shared crews/yard bottleneck. Warning: Weekly demand exceeds named capacity or coating/NDT queue grows. Owner: Construction. Response: Resource loading, workfront sequencing and priced additional capacity. Gate: Escalate before lookahead freeze.

R07 — Piping/platform movement mismatch. Warning: Stress/support reactions not accepted by structural team. Owner: Piping/Structural. Response: Iterate movement/stiffness and supporting deck checks. Gate: Bearing/service design release held.

R08 — Live-service loss or unsafe demolition. Warning: Isolation/bypass/temporary support not accepted. Owner: Operations. Response: Integrated SIMOPS/transfer and staged stability review. Gate: Stop affected work; complete controlled readiness.

R09 — NDT/coating rework or curing delays. Warning: Rejects, environment or DFT outside approved procedure. Owner: QA. Response: Qualify process, monitor environment, inspect and repair. Gate: No loadout before release.

R10 — Funding shortfall. Warning: Unpriced charters, unverified receipts or late certification. Owner: Finance. Response: Priced estimate, full-life cash, committed facility and scenario stress. Gate: Spending within authorised funded ceiling.

R11 — Measurement/payment rejection. Warning: No accepted e-MB/SES or amended milestone evidence. Owner: Controls. Response: Contemporaneous measurements and invoice reconciliation. Gate: Correct records before billing claim.

R12 — Emergency/weather disruption. Warning: Forecast reaches abort thresholds or vessel critical system fails. Owner: Marine/HSE. Response: Refuge/evacuation plan, drills, stop/abort authority and reserve strategy. Gate: Record decision and personnel accountability.

Risk likelihood/cost/schedule impacts remain unquantified; no probability orP80 claim is made. Use project-specific operating/abort limits and maintain actual weather, vessel/master decisions, standby causes and personnel accountability. Evaluate DGS/company audit provisions at mobilisation date. HSEand QAauthorities retain stop-work authority; management acceleration does not override the release gates.

Turnover is bridge/system based: structural/bearing inspection and accepted dimensions, weld/NDT/coating/material records, service reinstatement/tests and operations acceptance, scrap transport/yard receipt, native/as-built/vendor dossiers and punch closure. Commissioning is scoped to affected bridge services and interfaces; generic platform specifications do not automatically add full new process systems. Separate structural/mechanical completion, ready-for-service, company handover, final account and continuing warranty/security/integrity-pact obligations.

10. First 90 days and issue acceptance

Relative management targets areD1 contract/spending basis, D7 named organisation, D15 capacity/supplier offers, D30 survey/basis and validated scheduling plan, D45 first independently acceptable release where achievable, D60 funded procurement/released production, D75 measured productivity/marine assurance andD90 integrated feasibility decision. These targets are not represented as achieved. P14 base engineering release is later thanD45; therefore early-release objectives need a justified alternative sequence and capacity decision.

Accept the management baseline only after the actual start and controlling contract revisions are known, scope/MTO discrepancies resolved or explicitly dispositioned, design release logic accepted, supplier dates and resources evidenced, seasonal feasibility evaluated, control budget/cash funded and approvals signed. Until then useP14 as the single development-control index with versioned updates. Keep source issuesP8–P13 for audit references; retire conflicting working schedules only after an authorised baseline decision.

11. Deliverables and limits of completion

Delivered: this execution manual inWord/PDF; integratedExcelcontrol workbook; 285 activity CSV; readable bridge/phase schedule PDF; deterministic scenario data/scripts and source coverage inventories. Registers provide structured execution requirements, evidence and actions. The requested project-management development work is carried forward comprehensively; actual engineering certification, approved AFCs, orders/charters, fabrication, installation, testing, commissioning and accepted project completion are not performed by preparing this package.

No more missing-data approval question is needed to issue these reviewable documents. Subsequent commitments and construction releases must use the authorised contractual/technical decision process and real evidence. User-supplied files are treated as references; no external emails, portal submissions, declarations or orders were executed.